



EU Cloud Sovereignty Framework

Companion Guide for VMware Cloud Foundation 9.1

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Table of Contents

Disclaimer.....3

Version.....4

Introduction.....4

Third Party Identifiers and Mappings4

Coverage Levels4

Controls..... 6

 Control SOV-1..... 6

 Control SOV-2..... 7

 Control SOV-3..... 11

 Control SOV-4..... 18

 Control SOV-5..... 22

 Control SOV-6..... 23

 Control SOV-7..... 26

 Control SOV-8..... 32

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Version

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Introduction

This document describes how VMware Cloud Foundation (VCF) 9.1 and the separately licensed advanced services assessed alongside it, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, and VMware Avi Load Balancer, can support organizations addressing the sovereignty objectives of the European Commission's Cloud Sovereignty Framework. The framework, published by the Directorate-General for Digital Services (version 1.2.1, October 2025), defines eight Sovereignty Objectives covering strategic, legal and jurisdictional, data and AI, operational, supply chain, technology, security and compliance, and environmental sustainability dimensions, assessed through Sovereignty Effectiveness Assurance Levels (SEAL-0 through SEAL-4) in EU cloud procurement. The framework assesses cloud service providers; this guide describes how an organization operating VMware Cloud Foundation, such as a European cloud provider or a public-sector private cloud, can draw on platform capabilities when addressing each objective. The framework has no SCF crosswalk; the objective-to-SCF mapping in this guide was authored for the VCF assessment. Each assessed product is reported separately for every control, with VCF covering the full platform, including its consumption services and Private AI Services capabilities. The responsibility for demonstrating sovereignty assurance levels in a procurement remains with the tendering cloud service provider.

https://commission.europa.eu/document/download/09579818-64a6-4dd5-9577-446ab6219113_en

This guidance evolves. Please check the current revision at: <https://bcm.tech/vcf-compliance>

Third Party Identifiers and Mappings

This document includes regulatory compliance and security control identifiers from external sources as a convenience to end users. This does not constitute endorsement, in either direction.

There is not a one-to-one mapping of product capabilities to third-party controls. A product capability, or set of capabilities, may be applicable to multiple controls. Conversely, a control may be satisfied with the use of multiple capabilities.

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<https://securecontrolsframework.com/>

Coverage Levels

Each control entry below carries the framework control text, the SCF control or controls it maps to, and an Aggregate Coverage line showing the strongest position any one of the assessed products holds for that control. Products that meet or contribute to the control follow as subsections, each with its own coverage level and response. Products the control does not apply to are listed together at the end of the entry, with a brief explanation. The coverage levels are:

- **Meets** -- the product directly provides the capability described in the control. No additional organizational process, external product, or human activity is needed beyond configuring the product itself.
- **Contributes** -- the product provides technical capabilities that help satisfy the control, but the control also requires something outside the product to be fully met, such as an organizational process, human activity, or an additional tool.
- **Not Applicable** -- the control does not apply to the product. Physical facility controls, staffing decisions, organizational governance processes, and domains the product does not operate in fall here.
- **Not Supported** -- the control is relevant to the product's domain (infrastructure, networking, security), but the product lacks the specific capability the control requires.

- **Not Assessed** -- the control has not yet been evaluated against the product.

Controls

Control SOV-1

> Strategic Sovereignty: Strategic sovereignty captures the degree to which the services of a cloud provider (or technology actor) are anchored within the European Union legal, financial, and industrial ecosystem. It assesses ownership stability, governance influence, and alignment with EU strategic priorities. Contributing factors: Ensuring that bodies having decisive authority over your services are located within EU jurisdiction; Evaluating the assurances against change of control; Degree to which the provider relies on financing coming from EU sources; Extent of investment, jobs, and value creation within EU; Involvement in EU initiatives, Consistency with digital, green, and industrial sovereignty objectives defined at EU level; Ability to sustain secure operations against requests to cease or suspend the service, or if vendor support is withdrawn or disrupted.

SCF Controls: BCD-01, BCD-14, GOV-12, TDA-20.3, TPM-04

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control SOV-2

> Legal & Jurisdictional Sovereignty: Legal & Jurisdictional sovereignty evaluates the legal environment, exposure to foreign authority, and enforceability of rights that govern the services of a technology provider. It determines the extent to which the services are anchored in European jurisdiction and insulated from external legal claims. Contributing factors: The national legal system governing the provider's operations and contracts; Degree of exposure to non-EU laws with cross-border reach (e.g., US CLOUD Act, Chinese Cybersecurity Law); Existence of legal, contractual, or technical channels through which non-EU authorities could compel access to data or systems; Applicability of international regimes, which may restrict usage or transfer; Location of intellectual property creation, registration, and development (EU vs. third countries), legal jurisdiction where IP is created and developed.

SCF Controls: CLD-09, CPL-01, DCH-14.2, DCH-26, TPM-04.4

Aggregate Coverage: Contributes

VCF (Contributes)

VCF provides several layers of technical capability that help organizations manage data placement and enforce geographic constraints, which is relevant when addressing digital sovereignty requirements that mandate data localization within specific jurisdictions.

VCF Automation organizes infrastructure into regions, and a region quota defines the compute and storage quota that an organization receives against a given region. The provider administrator decides from which regions to allocate quota to each organization, giving centralized control over where workloads and their associated data can be provisioned. Image mappings are also region-aware, with available images determined by the selected region. This region-based architecture allows administrators to constrain resource provisioning to specific geographic locations, supporting data residency requirements. Cloud template location constraints can be applied to both Machine and Disk resources to separate disk placement across locations, giving fine-grained control over where compute and storage resources are placed within a deployment.

Constraint tags in cloud templates further refine placement behavior. Tags added to projects and cloud templates function as constraint tags when matched against capability tags on infrastructure resources, profiles, and cloud zones. Project-level constraint tags take precedence over cloud template constraint tags at deployment time, allowing cloud administrators to enforce governance rules centrally. For example, if an administrator creates a location:london tag on a project but a developer places a location:boston tag on a cloud template, the project tag takes precedence and the resource is deployed to infrastructure containing the location:london tag. Constraint tags can also use variables to populate tags at request time, enabling flexible location selection within administrator-defined boundaries.

vSphere VM Storage Policies provide an additional placement control mechanism at the virtual machine level. Tag-based rules within VM Storage Policies allow administrators to filter datastores by geographic location, including or excluding datastores based on location-encoded tags. Datastores not represented by storage providers can be tagged within VM Storage Policies to communicate geographic properties that are not available through the storage provider interface, allowing geographic placement constraints to be enforced at the storage policy layer.

Policy as Code in VCF Automation enables organization administrators to enforce instance-centric policies for Infrastructure as a Service resources from a central location. These policies can restrict where workloads are deployed and what resources they can consume, providing a programmatic mechanism for enforcing placement constraints that align with data localization obligations.

For data protection at rest, vSAN provides cluster-based data-at-rest encryption. vSAN encryption requires a Key Management Server (KMS) and supports integration with a KMS cluster for encryption key management. vSphere Native Key Provider is also available as an alternative for encrypting virtual machines in vSAN clusters. The separation of encrypted data from key material allows organizations to place encrypted workloads in a required jurisdiction while maintaining control of encryption keys in a separate, trusted location, limiting the exposure of data at rest to enforcement actions in a single jurisdiction.

For multi-site deployments, NSX Federation provides a topology where a Global Manager coordinates with Local Managers at each site. On a Local Manager, administrators can see local objects and any global objects that apply to that location, while the Location Manager monitors communication details between Local Manager and Global Manager. Security objects in NSX Federation have a region that can be Location (automatically created for each location with that location's span), Global (spanning all available locations), or Custom Region (spanning a subset of available locations). This architecture gives operators location-aware visibility and control over both networking and security configuration at each site, which is relevant when different jurisdictions impose different requirements on network traffic and data flows.

VCF Operations provides location-awareness features that help track where infrastructure resides. A GEO Location tag type allows tag values to be identified with a location on a world map, and administrators can select a physical location on a map for each data center with search and zoom capabilities. Objects assigned a GEO Location tag appear on a geographical map in VCF Operations, and the Geo Widget shows the health of each object at each location. These tagging and mapping features support inventory and compliance tracking for geographically distributed infrastructure.

Organizations should also be aware that cross-zonal datastores typically replicate data across zones. When data sovereignty requirements prohibit cross-border replication, storage architecture decisions must account for this behavior to avoid unintended data movement outside permitted jurisdictions.

Defining organizational data residency policies, mapping regulatory requirements to specific jurisdictions, managing contractual obligations around data localization, and determining which jurisdictions apply to which workloads are organizational responsibilities outside VCF's scope.

VMware vDefend (Contributes)

VMware vDefend (vDefend) contributes to constraining the impact of digital sovereignty laws through GeoIP visibility, IDS/IPS-based geographic detection, distributed firewall scoping, and federated policy management at the network layer.

vDefend's Security Services Platform includes GeoIP Monitoring, which displays a map of network traffic to and from every country in the world. This capability requires the Security Services Platform to be deployed with the metrics feature enabled and gives administrators visibility into geographic traffic patterns across the environment. vDefend IDS/IPS signature rules support geoip matching on source IP addresses, destination IP addresses, or both, providing a detection mechanism that identifies traffic associated with specific countries within east-west distributed traffic flows. This allows administrators to detect and respond to data flows that cross jurisdictional boundaries where policy requires them to remain local.

vDefend supports policy scoping across global, regional, and local spans in a federated deployment. Security Services Platform regions can be associated with Security and Network policies to control policy scope across multiple locations. DFW policies applied globally allow sources or destinations to be any object without span restrictions, while the Applied To field scopes enforcement to specific zones, tenants, or applications without affecting policies defined for other workloads. This regional policy architecture allows organizations to define distinct security controls for workloads in different jurisdictions while maintaining a consistent baseline across sites. In a federated environment, malicious traffic can be blocked across Local Managers from the Global Manager, providing centralized enforcement across geographically distributed sites.

These network-layer capabilities complement the broader VCF data sovereignty mechanisms, including encryption and workload placement constraints, by providing traffic-level controls and visibility that support geographic boundary enforcement. Defining organizational data residency policies, mapping regulatory requirements to specific network policies, and managing contractual obligations around data localization remain organizational responsibilities outside vDefend's scope.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) supports data localization through customer-controlled site pairing and explicit replication targeting. PNR operates within customer-owned and customer-managed infrastructure, replicating only between explicitly paired VMware vCenter sites that an administrator configures. Protection groups define which virtual machines replicate to which recovery site, and replication between sites requires deliberate administrator action.

PNR supports deployment across multiple geographic regions, with PNR Server and vCenter instances at each site running in locale-specific configurations. Documented topologies include a Japanese locale site paired with a United States English

locale site, with each PNR Server extending the vCenter Appliance instance at its own site. This allows organizations to align recovery infrastructure with jurisdictional boundaries and keep replicated workloads within specified regions.

Site pairing is configured between two specific vCenter instances, each running on infrastructure the customer physically controls. Inventory mappings between sites are configured manually, mapping networks, folders, compute resources, and datastores from the protected site to corresponding objects on the recovery site. An optional reverse mapping capability creates the equivalent mappings from the remote site back to the local site to support re-protection. Manual storage policy mapping allows multiple storage policies on the local site to be mapped to a single storage policy on the remote site. These administrator-driven mappings keep placement decisions for protected and recovered workloads within the customer's control.

For PNR deployments that integrate with VCF Automation, a content library must be configured and available for upload in each region where PNR is being deployed, reinforcing the per-region scoping of recovery infrastructure. The VCF Operations Orchestrator plug-in for PNR provides workflows for listing and managing network, folder, and resource mappings on a per-site basis, supporting programmatic verification that data placement aligns with jurisdictional requirements.

Cyber Recovery, the ransomware recovery capability included in PNR, uses data seeds, which are pre-populated datasets that jumpstart replication in a new location by providing the target with an initial copy of source data. Protection groups allow administrators to selectively replicate workloads, so data subject to localization requirements can be excluded from cross-border replication or directed only to recovery sites within the same jurisdiction.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) contributes to this control through role-based access controls, identity integration, and connection-level authorization mechanisms that restrict data transfer operations to authenticated and authorized entities.

DSM provides two platform-level roles: DSM Admin and DSM User. These roles govern what operations a user may perform, including provisioning databases and establishing replication relationships between instances. Users can be sourced from LDAP or LDAPS identity providers, or maintained as local accounts. When directory service integration is configured, DSM uses the configured directory service for both authentication and authorization. Administrators import LDAP users through the DSM Admin UI and assign them the DSM Admin or DSM User role. MySQL database instances can further delegate authentication to Active Directory via LDAPS. SQL Server databases support Windows Principal authentication, allowing users to authenticate with Active Directory credentials; administrators can optionally configure privileged Active Directory account credentials to activate automated Kerberos integration. Role assignment determines which users are permitted to initiate operations that involve transferring data, such as establishing replication or running migrations.

For client applications and services connecting to DSM-managed database clusters, DSM enforces certificate-based trust requirements before connections can be established. Custom trusted certificates used in DSM must comply with FIPS requirements. Database clusters issue certificates for client connectivity and monitoring connections. PostgreSQL databases support host-based authentication (HBA) configuration, with the default configuration requiring SCRAM-SHA-256 authentication for service accounts including `postgres_exporter` and `pgadmin`. When LDAPS users access PostgreSQL databases, administrators configure specific privilege grants, including database and schema-level access, before those users can perform operations. These mechanisms control which client systems and identities may establish connections and exchange data with the managed databases.

Data transfer operations between DSM database instances carry additional authorization requirements tied to role and identity verification. During PostgreSQL database migration, DSM requires TLS verification between source and target clusters. For multi-site PostgreSQL replication, users with the DSM User role must manually establish trust between primary and secondary instances before replication proceeds; users with the DSM Admin role have trust established automatically. This role distinction governs which identities may initiate and complete cross-instance data transfers. MySQL database migration to DSM requires creation of a dedicated user account with minimum privileges of `BACKUP_ADMIN`, `REPLICATION_SLAVE`, `SELECT`, and `INSERT` on the `mysql` schema, restricting the migration account to only those permissions the operation requires. DSM also verifies network access to the source database and validates SSL trust during MySQL replication operations.

For data forwarded to external systems, DSM requires authenticated connections before transfer proceeds. Metrics forwarding supports Mutual TLS (mTLS) authentication, configured by enabling the Client Certificate Authentication option and supplying the client certificate and private key. Forwarding to other metrics targets requires verification of server address, port, authentication type, credentials, and certificates before the forwarding connection is accepted. Webhook notification endpoints support basic authentication with Authorization User and Password credentials, bearer token authentication via Authorization header, or authorization embedded in the webhook URL for Slack-compatible endpoints. For self-service database consumption, the Consumption Operator requires a TLS certificate for secure communication with the DSM provider, and an authentication secret containing the credentials needed to connect to the DSM provider must be created and configured before the operator can connect.

DSM's authorization mechanisms primarily govern user-initiated operations and connection-level access within the database service layer. Verifying authorizations for all Technology Assets, Applications and/or Services transferring data across interconnected systems requires organizational data governance processes that extend beyond DSM's scope. Organizations must define which services are authorized to connect to DSM-managed databases, maintain records of authorized interconnections, and audit application-layer data flows independently of the platform controls DSM provides.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer provides several application-layer mechanisms that support data localization requirements by controlling where application traffic is directed and which clients can access specific services.

The Global Server Load Balancing (GSLB) feature in Avi includes a geolocation-based load balancing algorithm that directs client requests to region-specific application deployments based on the geographic origin of the request. Avi maintains a geolocation database on its Controllers, which can be loaded from file objects of type GSLB Geo DB and used as a Geo profile for the geolocation-based algorithm. The GSLB geolocation database supports multiple input files with a priority mechanism that determines which source takes precedence when multiple databases provide data for the same IP address. Geolocation database records include IP address range start and end, latitude, longitude, and an optional name field representing city, region, and country in a hierarchical format.

Avi's DNS Policy feature provides granular DNS-level traffic steering and filtering based on geographic criteria. DNS policies can match on client IP addresses, protocol, query name, query type, or geographic location, and apply actions including allowing or blocking DNS queries, rate limiting, or selecting a GSLB site. DNS Policy rules support geolocation-based matching criteria and GSLB group selection actions, enabling country- and region-level enforcement directly in the DNS resolution path. DNS Policy can block DNS queries from a particular geographic area by configuring a client IP match rule with IP addresses associated with that area and setting the rule action to drop.

For custom programmable logic, Avi DataScript provides the `avi.utils.get_geo_from_ip` function, which retrieves geolocation information including autonomous system number, country code, country name, continent name, continent code, and geographic coordinates (latitude, longitude, region, and city) for a given IP address. Avi DNS DataScript also supports geo-based DNS response routing, allowing different A record responses to be returned based on the client's country location. This enables organizations to implement dynamic, policy-driven restrictions on which clients can reach specific application endpoints.

Avi's Web Application Firewall (WAF) integrates with the geolocation database. The WAF Policy can be configured with a Geo Location Mapping Database, and the Avi Service Engine supports geolocation database usage in HTTP and network policies at region and city levels, not limited to country-level granularity. Geo-mapping attributes including `ATTRIBUTE_COUNTRY_CODE` and `ATTRIBUTE_ISP_NAME` can be referenced in security policies to apply geo-based access controls. The Geo-DB mapping feature can group geographic regions such as APAC countries or embargoed countries for use in policy matching and enforcement across multiple rules.

These capabilities operate at the application delivery layer. Data localization governance also requires organizational policies governing data classification, workload placement decisions at the VCF platform level, and procedures for responding to enforcement actions under applicable laws.

Control SOV-3

> Data & AI Sovereignty: Data & AI sovereignty focuses on the protection, control, and independence of data assets and AI services within the EU. It addresses how data is secured, where it is processed, and the degree of autonomy customers retain over AI capabilities. Contributing factors: Ensuring that only the customer, not the provider, has effective control over cryptographic access to their data; Visibility into when, where, and by whom data is accessed, including auditability of AI model usage, mechanisms guaranteeing irreversible removal of data, with verifiable evidence; Strict confinement of storage and processing to European jurisdictions, with no fallback to third countries; Extent to which AI models and data pipelines are developed, trained, hosted, and governed under EU control, minimizing dependency on non-EU technology stacks.

SCF Controls: AAT-10.17, AAT-10.19, CLD-09, CRY-01, CRY-09, CRY-09.4, CRY-09.7, DCH-09, DCH-09.1, DCH-26, IAC-21.4, MON-02.7, MON-03.2, RSK-09.2

Aggregate Coverage: Contributes

VCF (Contributes)

VCF implements cryptographic protections across its component stack using publicly standardized, validated cryptographic modules and algorithms.

Multiple VCF components rely on cryptographic modules validated through the NIST Cryptographic Module Validation Program (CMVP) to FIPS 140-2 and FIPS 140-3 standards. NSX is configured to use FIPS 140-3 validated cryptographic modules by default, and NSX Manager uses FIPS 140 approved algorithms for authentication to its cryptographic module. Specific NSX validated modules include IKE with Rambus Safezone 2.0 (Certificate #4898), VMware OpenSSL 3.0.9 (Certificate #4861), VMware BouncyCastle 2.0.0 (Certificate #4986), and VMware BoringCrypto 6.0 (Certificate #4694). VCF Automation uses the BouncyCastle FIPS 2.0.0 cryptographic module, which holds FIPS 140-3 Certificate #4743. VCF Operations for Networks uses BC-FJA (Bouncy Castle FIPS Java API) version 2.0.0 with FIPS 140-2 Certificate #37009 as its validated cryptographic module. This consistent use of FIPS-validated modules across the platform means that organizations deploying VCF in regulated environments can rely on cryptographic implementations that have undergone independent validation against well-known public standards. The VMware Kubernetes Service (VKS) consumption layer also supports FIPS mode: VKS ClusterClass configurations include a `fips.enabled` variable that, when set, restricts cryptographic operations on cluster nodes to approved algorithm implementations. FIPS-enabled Kubernetes release images, identified by a `-fips` suffix in the release reference name, are available for organizations that require FIPS-validated cryptography across their containerized workloads.

At the protocol and algorithm level, VCF enforces modern cryptographic standards throughout. Starting in vSphere 9.0, all TLS profiles are FIPS compliant; the COMPATIBLE profile supports modern TLS cipher suites with standardized elliptic curves. Monitoring infrastructure such as Telegraf enforces a modern TLS minimum version for secure communication. The NSX L2 VPN FIPS compliance suite uses AES-GCM encryption with Perfect Forward Secrecy and Diffie-Hellman Group 20. NSX also supports ECDSA (Elliptic Curve Digital Signature Algorithm) encryption for environments with high-assurance cryptographic requirements. VCF Operations for Networks supports several encryption algorithms and ciphers for its data sources. VCF Operations provides a FIPS Security Mode that, when enabled, applies additional algorithm and cipher prerequisites across the management domain. All system management network communications are encrypted by default using standardized cipher suites. At the VKS layer, Kubernetes control plane components expose parameters for restricting permitted TLS configurations: the `kube-apiserver --tls-cipher-suites` parameter, the `kubelet tlsCipherSuites` configuration field, and the equivalent parameter on the `kube-scheduler` each restrict the set of permitted cipher suites, allowing administrators to exclude deprecated or weak algorithms. The `kubelet tlsMinVersion` field and the `--tls-min-version` parameter on the `kube-apiserver` enforce a minimum TLS protocol version floor for inbound connections. The Antrea networking component and vSphere Cloud Provider Interface on VKS clusters also support configurable TLS cipher suites for component-to-component communication. Kubernetes hardening guidance identifies specific preferred cipher suites and calls out cipher suite configuration on the `kube-scheduler` as a required hardening step. VKS clusters support the Istio service mesh package, which provides mutual TLS between workloads within the mesh; the `meshMTLS.minProtocolVersion` setting controls the minimum protocol version for inter-service communication, defaulting to TLS 1.2 with support for configuring TLS 1.3.

Data-in-transit protections using TLS have customizable PKI keypair and certificate options. These range from fully automated within VCF to semi-automated using certificates from enterprise PKI infrastructure or fully custom certificates installed manually. Automated certificate issuance is supported with Microsoft Active Directory Certificate Services. Trust establishment uses standard PKI management with multiple certificate options including Root CA certificates, CSR-based certificates, and direct certificate uploads. VCF Operations provides centralized certificate lifecycle management for all VCF components, enabling automated certificate renewal, monitoring, and replacement across the infrastructure. Organizations should be aware that if self-signed certificates and private keys are generated using OpenSSL, those certificates and private keys are not FIPS-compatible. Environments operating under FIPS requirements should use certificate authorities and key generation methods that produce FIPS-compliant artifacts. For FIPS environments, certificates should use key sizes of 2048 or 3072 bits; key sizes greater than 3072 bits have not been tested with FIPS. At the VKS layer, the cert-manager package manages the lifecycle of certificates issued to workloads and ingress endpoints. VKS clusters maintain TLS certificates for component-to-component communication across their control and data planes. The Contour ingress package supports minimum TLS version configuration through the `tls.minimum-protocol-version` parameter. Harbor, when deployed as a Supervisor Service, accepts custom TLS certificate configuration through the `tlsCertificate.tls.crt` and `tlsCertificate.tls.key` fields, allowing organizations to supply certificates from their enterprise certificate authority rather than relying on self-signed certificates. The Supervisor secret injection agent automatically configures mutual TLS for communication with the vault-agent-injector service using auto-generated certificates.

Data-at-rest encryption uses vSAN cluster-based encryption or VM Encryption on non-vSAN storage. vSAN encryption uses AES-NI CPU offloading for efficient processing, and VMware ESX hosts encrypt vSAN disk data using the industry standard AES-256 XTS mode. Encryption keys are managed through KMIP 1.1-compatible external Key Management Servers (KMS) or vSphere Native Key Provider (NKP), which is included in all vSphere editions and does not require an external key server. VCF implements a three-element domain of trust comprising the key provider, VMware vCenter, and vSAN hosts. Key Encryption Keys (KEK) wrap Disk Encryption Keys (DEK) for secure key management; ESX hosts use the KEK to encrypt their internal keys and store only the encrypted internal keys on disk, not the KEK itself.

VCF Standard Key Providers have key wrapping capabilities that let organizations store a single wrapping key in their KMS, protecting multiple Key Encryption Keys within the VCF environment. This creates a three-tier key hierarchy: the Data Encryption Key encrypts object data, the Key Encryption Key protects the DEK, and the wrapping key protects the KEK. This approach reduces the number of keys stored in external KMS systems while maintaining cryptographic separation. Key wrapping includes automatic key rotation with configurable intervals from 30 days to 10 years.

With key persistence enabled, ESX hosts can persist encryption keys in the Trusted Platform Module (TPM) across reboots, allowing encryption operations to continue when the key server is unavailable. Each ESX host obtains the encryption keys initially and retains them in its key cache; if the ESX host has a TPM, the encryption keys are persisted in the TPM across reboots. Key persistence can be enabled using the `esxcli system security keypersistence enable` command.

VCF supports guest-level security features including virtual Trusted Platform Module (vTPM) 2.0 for VMs. This enables in-guest encryption capabilities such as Windows BitLocker, Linux disk encryption, and other operating system-native security features. Secure Boot provides UEFI verification to help ensure only signed and trusted code runs during the VM boot process. These features work independently of or in combination with infrastructure-level encryption, allowing organizations to implement defense-in-depth strategies.

vSAN offers Data-in-Transit (DIT) encryption as a cluster-wide setting that encrypts all data and metadata traffic as it transits across hosts. For vSAN Original Storage Architecture (OSA), data-at-rest encryption occurs at the Local Object Manager (LSOM) layer on the target host, making DIT encryption strongly recommended to restrict network-level interception. For vSAN Express Storage Architecture (ESA) in HCI mode, data is encrypted at the VM host before network transmission, making DIT encryption optional based on compliance requirements and risk tolerance. DIT encryption operates without requiring external key management servers and provides independent key rotation from data-at-rest encryption.

Encrypted vSphere vMotion secures confidentiality, integrity, and authenticity of data transferred during VM migrations. For encrypted VMs, vMotion always uses encryption. For unencrypted VMs, encrypted vMotion can be set to Disabled, Opportunistic (default), or Required. Cross-vCenter vMotion is supported when using shared key providers or backup/import

of Native Key Provider. VCF uses one-time keys (nonces) for each migration session so that captured traffic cannot be replayed or decrypted after migration completes.

Memory Tiering encryption provides protection for tiered memory without requiring external key management. VCF generates random 256-bit AES-XTS keys at the kernel level during VM power-on, unique to each VM instance or per host depending on configuration.

VM Encryption provides protection across all datastore types including vSAN, iSCSI, NFS, and Fibre Channel. Disk encryption uses XTS-AES-256 keys as the disk encryption key. This encryption occurs before data reaches the storage layer, providing both data-at-rest and data-in-transit protection.

Kubernetes on VKS supports encryption at rest for data stored within the control plane. Operators configure the encryption provider and manage key access for control plane secrets; when relying on a managed key provider, administrators retain responsibility for access controls over the managed key material. The VCF Automation consumption layer includes VCF Encryption Management, which grants organization administrators the ability to use encryption keys from their own key providers for encryption of virtual machines and disks in their VCF environment, extending the platform's cryptographic key management to tenant-level operations.

For storage protocols, VCF provides varying levels of native encryption support. iSCSI includes CHAP for session authentication. Fibre Channel Gen 7 supports end-to-end encryption with secure HBAs. For NFS datastores, version 4.1 supports Kerberos authentication with three security levels: krb5 (authentication only), krb5i (authentication plus integrity checking), and krb5p (authentication plus privacy with full encryption).

ESX stores configuration data in encrypted form on boot devices. When the host has a TPM 2.0 enabled and configured, the configuration encryption key is stored in the TPM. The encryption mode can be configured using the `esxcli system settings encryption set` command with the `--mode=TPM` parameter.

Cryptographic operations are controlled through fine-grained privileges in vCenter. Only users assigned the Cryptographic Operations privileges can perform cryptographic operations. A dedicated No Cryptography Administrator role enables standard VM and infrastructure management while restricting unauthorized encryption changes. This role can be cloned and customized to grant only specific Cryptographic Operations privileges, such as allowing encryption but not decryption, supporting the principle of least privilege for encryption tasks. The `CryptoManager` and `CryptoManagerKmp` managed objects provide programmatic interfaces for handling cryptographic keys and integrating with key management server infrastructure. vSAN encryption key management requires specific permissions including `Cryptographer.ManageEncryptionPolicy`, `Cryptographer.ManageKeyServers`, and `Cryptographer.ManageKeys`.

vSAN includes secure disk wipe capabilities for proper media sanitization when decommissioning storage, supporting the Erase disks before use option during encryption configuration and disk management operations.

Defining the organization's cryptographic policy (algorithm selection, key strength, rotation schedules, KMS architecture, FIPS-mode activation decisions) and the key lifecycle management standards that govern key creation, distribution, storage, archive, and destruction sit with the organization above the platform. VCF supplies the validated cryptographic modules and protocol enforcement; the organizational cryptographic governance program determines how those mechanisms apply across the environment. Security Technical Implementation Guides (STIGs) for Supervisor and VKS releases are available to guide cryptographic hardening of the consumption layer.

VMware Private AI Services (PAIS) provides cryptographic controls across its AI infrastructure components, building on the cryptographic foundation of VCF.

GPU-accelerated VMware Kubernetes Service (VKS) clusters deployed for PAIS workloads use a FIPS-enabled Kubernetes runtime. The VKS cluster topology version applied to both control plane and worker deployments uses a FIPS-enabled VKR build running on Ubuntu 24.04, so AI workloads on PAIS VKS clusters operate within a FIPS-compliant Kubernetes runtime.

PAIS endpoints use TLS for service communication. The `PAISConfiguration` custom resource supports configurable CA bundle references through `spec.clientTls.caBundleRefs` for OIDC provider and Harbor registry authentication. The Supervisor

endpoint communication for PAIS uses HTTPS on port 6443 with certificate-based authentication. The PAIS Prometheus metrics endpoint uses self-signed certificate authentication with TLS, where the CA certificate is retrieved from the `pais-ingress-default Secret` in the PAIS namespace and configured in base64-decoded form. Prometheus TLS client authentication and certificate validation are off by default; auditors should verify that client authentication is enabled in production deployments where mutual certificate validation is required.

PAIS requires CA trust bundles to be configured for all external HTTPS connections, including OIDC providers, Harbor registries, content management systems used as data sources in the Data Indexing and Retrieval service (such as Confluence or SharePoint), MCP servers that provide tools to agents, and the pgvector database server used for vector storage. Trust bundle material is obtained from each external system's administrator either as a certificate file or as a ConfigMap stored in the namespace where PAIS is deployed, and PAIS recommends maintaining separate trust bundles per application with periodic updates before certificate expiration. Administrators add trusted certificates through the VCF Automation Provider Management UI under Certificate Management and Trusted Certificates. The issuer certificate of the Harbor registry must also be added to the certificate trust store on Deep Learning VMs that pull models from the registry. When connecting to an MCP server over HTTPS, the MCP server issuer certificate must be added as a CA trust bundle.

The underlying cryptographic algorithms, cipher suites, and platform-level key management are provided by VCF and VKS. Organizations are responsible for selecting and enforcing approved cryptographic standards at the VCF and infrastructure level, while PAIS provides the certificate management and trust chain configuration mechanisms specific to AI service endpoints.

VMware vDefend (Contributes)

VMware vDefend contributes to the implementation of cryptographic protection controls by using known public standards and trusted cryptographic technologies across its component stack for securing inter-component communications and providing cryptographic inspection capabilities for network traffic.

Security Services Platform (SSP), the control-plane component of vDefend, secures communications between vDefend product components using certificates and current TLS protocol versions with approved cipher suites. SSP and the SSP Installer deploy with the most current TLS version by default and also support the prior major TLS version. The TLS versions and cipher suites allowed on SSP Installer and SSP are fixed and cannot be modified, which enforces a consistent cryptographic baseline across deployments.

SSP components, including the SSP Installer and the NDR Sensor for vDefend, implement FIPS 140-2 and FIPS 140-3 cryptographic module standards. SSP uses the VMware BouncyCastle Module (NIST CMVP Certificate #4943) and the VMware OpenSSL FIPS provider Object Module (NIST CMVP Certificate #4861) for its cryptographic operations. SSP actively enforces cryptographic hygiene by rejecting any certificate or TLS configuration that uses deprecated algorithms, short keys, or legacy protocol versions, returning a validation error rather than permitting degraded connections.

Certificate management within SSP provides lifecycle controls including certificate export, certificate signing request (CSR) generation, and CA-signed certificate replacement. SSP monitors and manages several certificate types used to identify client and host machines, establish trust between components, secure communications, and control access levels. Administrators can retrieve and validate LDAP certificates using standard tools to establish secure LDAPS connections for platform authentication. These capabilities allow organizations to integrate vDefend into their broader PKI infrastructure and follow standard certificate lifecycle practices.

vDefend provides TLS Inspection capabilities that allow the firewall to decrypt, inspect, and re-encrypt TLS-protected traffic. Without TLS Inspection, encrypted traffic cannot be examined or enforced against security policies even when other advanced security features are enabled. With TLS Inspection enabled, vDefend gains visibility into encrypted traffic flows, supporting more effective access control and threat detection. TLS Inspection also supports importing or generating trusted and untrusted proxy certificate authorities for external decryption profiles, giving administrators control over the CA chain used in inspection.

TLS Inspection includes a Crypto Enforcement option within decryption action profiles that allows administrators to set minimum and maximum protocol version and cipher suite constraints for both client and server connections. Profiles can be

configured in Transparent or Enforce modes; the Enforce mode requires compliance with the configured cryptographic constraints.

vDefend IDS/IPS signature rules extend cryptographic visibility by supporting protocol version matching and certificate Subject field matching, enabling enforcement decisions based on certificate metadata in encrypted traffic. IDS/IPS rules also support storing TLS/SSL certificates to disk for forensic analysis and threat investigation. SSP guidance recommends that organizations restrict or block protocols that carry data in cleartext and enforce minimum secure protocol versions as part of their network security practices.

vDefend does not itself provide the underlying key management infrastructure (KMS), platform-wide PKI lifecycle management, or the broader VCF platform's cryptographic controls; those responsibilities belong to VCF. However, vDefend's consistent use of approved cryptographic standards for its own operations and its ability to perform cryptographic inspection of network traffic contribute to an organization's overall implementation of cryptographic protection controls.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) implements cryptographic protections for disaster recovery workflows using known public standards and trusted cryptographic technologies.

PNR uses modern TLS by default for all connections to the PNR server. The minimum TLS protocol version is configurable through the `envoy-proxy-config.json` file by editing the `tls_minimum_protocol_version` parameter. When the minimum protocol version is modified, PNR requires that the change be applied consistently across all relevant configuration occurrences in the system. Administrators can verify the TLS version in use on the Site Recovery Manager component using standard OpenSSL tooling.

The PNR Appliance requires an SSL/TLS certificate as the endpoint certificate for all TLS connections to the PNR server. PNR supports custom certificates signed by a certificate authority as an alternative to self-signed certificates. Custom SSL/TLS server endpoint certificates must meet specific requirements, including x509v3 Extended Key Usage indicating TLS Web Server Authentication. The Appliance Management Interface allows administrators to import CA-signed server certificates and custom root CA certificates, and to replace the appliance certificate. PNR uses TLS certificates and private keys to protect network communication and establish authentication with other servers. Certificate verification error messages are logged in security-related log files, and operational log files do not contain sensitive information such as private keys or passwords.

For encrypted virtual machine replication, PNR requires that the recovery and protected sites use a common Key Management Server (KMS) or that the KMS clusters at both sites share common encryption keys. Replication of encrypted virtual machines also requires Secure LWD support to be available on the virtual machine. This supports data-at-rest encryption continuity for virtual machines through the recovery lifecycle.

Defining the organization's cryptographic policy, including KMS architecture decisions, minimum TLS version selection, certificate management standards, and key lifecycle practices, sits with the organization above the DR layer. PNR supplies the cryptographic mechanisms needed for DR-specific traffic and operations; the organizational cryptographic governance program determines how those mechanisms apply across the environment.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) implements cryptographic protections across multiple layers of its architecture, applying standard cryptographic technologies for data in transit, certificate trust management, and credential storage.

Starting with DSM, the platform enforces FIPS 140-3 compliant cryptographic standards and validates certificates during upload, rejecting unsupported algorithms or parameters. Only RSA and ECDSA public key algorithms are accepted for certificates. The Ed25519 elliptic curve signature algorithm is rejected because it does not comply with FIPS 140-3. Custom trusted certificates should use ECDSA P-256 or P-384 algorithms; RSA certificates must meet a minimum key size of 2048 bits per FIPS requirements. DSM's latest certificate validation mode enforces strict cryptographic and KeyUsage validations across all DSM endpoints, including the Provider VM and data service clusters.

For connections to database clusters, DSM requires TLS for all client traffic. PostgreSQL database instances enforce a minimum TLS protocol version through the `ssl_min_protocol_version` setting. MySQL enforces minimum TLS versions on both the main connection interface and the administrative interface; these settings are fixed and cannot be modified by users, which helps maintain a consistent cryptographic baseline across deployments. MySQL also exposes the `group_replication_recovery_tls_version` setting to specify the TLS protocol used during distributed recovery connections. DSM supports SQL Server clusters, which use TLS to encrypt communication between clients and the cluster; the `network.tlsProtocols` custom engine setting defines the allowed TLS versions for client connections, and a minimum TLS version must be included for the DSM operator to manage the cluster. DSM supports TLS verification during PostgreSQL database migration, covering replication traffic between source and target clusters.

DSM's certificate management capabilities support standard PKI practices. Administrators configure custom TLS certificates for database clusters through the DSM UI or API, providing certificates in PEM format with associated private keys and CA certificates. Supported private key algorithms are RSA and ECDSA with key sizes of 256 and 384 bits. When present, the X509v3 KeyUsage extension on a custom certificate must include the DigitalSignature capability; DSM also accepts certificates that omit the KeyUsage extension entirely, treating absent KeyUsage as permitting all usages in accordance with RFC 5280. To establish secure connections to database clusters and to the DSM Provider VM, clients add the CA to the operating system trust store or the trusted CA list. DSM's API provides a mechanism to retrieve the CA certificate from a custom certificate secret. The Consumption Operator, which enables self-service database provisioning, requires a TLS certificate for its communication with the DSM provider. Certificate management can also be handled through Kubernetes cert-manager, which manages various issuer types and outputs Kubernetes TLS Secrets in the format DSM requires.

DSM supports TLS for connections to external systems. When configuring log forwarding to an SMTP server, DSM supports Auto TLS to establish a secure connection. For private container registries, DSM supports CA certificate validation for untrusted registries.

For local user credential storage, DSM encrypts passwords using PGP symmetric encryption through the PGCrypto library, with dedicated encryption keys stored using the `PGCRYPTO_PASSWORD` and `PGCRYPTO_ALGORITHM` keys. MySQL has FIPS mode available on the client side through the `ssl_fips_mode` setting. PostgreSQL databases include the `pgcrypto` extension, which provides cryptographic functions that applications can use directly within the database. The DSM Provider Appliance includes virtual machine encryption, which activates when a Key Management Server is configured in the environment.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer provides configurable SSL/TLS profiles that control which cipher suites, protocol versions, and key exchange algorithms are permitted for application traffic termination. The SSL/TLS Profile, configured under Templates > Security > SSL/TLS Profile, contains the accepted protocol version list and a prioritized cipher suite list applied to virtual services. Administrators can create custom profiles or select from built-in templates; the documentation notes that the choice of accepted ciphers and protocol versions involves trade-offs between security, client compatibility, and computational expense. SSL profiles in 9.1 include a Post Quantum Cryptography section with support for modern cipher suites. Controller SSL/TLS certificates support elliptic curve (EC) or RSA algorithms, with EC recommended for new deployments.

For environments subject to cryptographic module validation requirements, Avi supports a FIPS mode backed by the OpenSSL FIPS Provider, which has been validated at FIPS 140-3 Security Level 1 and awarded Certificate #4985 by the Cryptographic Module Validation Program (CMVP). When FIPS mode is activated on the Controller cluster, Avi restricts cryptographic operations to FIPS-compliant algorithms only. FIPS 140-3 also requires stricter handling of sensitive security parameters including keys, authentication data, and random number generation. FIPS mode is enabled at the Controller cluster level through the Controller settings. Note that FIPS mode does not support SSL Client Hello events in Avi DataScript, which may affect certain scripted traffic inspection use cases.

Avi Controller allows administrators to restrict specific key exchange algorithms and ciphers for management session access. The Allowed Ciphers field restricts management session ciphers to a specified subset, and the `kex_algorithm_exclude` field under Controller Settings excludes specific key exchange methods for SSH-based administrative access. Internal

communications between the Avi Controller and Service Engines also use SSL/TLS, and custom certificates can be selected for this Secure Channel connection via the Access tab certificate field.

For certificate management, Avi integrates with Hardware Security Modules (HSMs), including the Thales Luna (formerly SafeNet Luna) HSM, for cryptographic key storage and management. The Controller is set up as an HSM client and can create keys and certificates directly on the HSM through client libraries, supporting high-availability HSM configurations. Automated certificate lifecycle management is available through Let's Encrypt integration using the ACME protocol. Avi's App Transport Security feature supports integration with the Venafi Trust Protection Platform for certificate administration across tenants, with the Trust Protection Platform pushing signed certificates and required chain certificates to Avi Load Balancer.

For Kubernetes deployments, the Avi Kubernetes Operator (AKO) supports PKIProfile-based certificate validation and trust management for backend services. The PKIProfile, configured through custom resources, enables secure communication with backend services through certificate-based authentication and encryption. In VCF environments, the Avi PKI Profile defines the list of trusted certificate authorities permitted to sign client certificates and can be used to implement Zero Trust security between servers and clients.

Data-at-rest encryption for Avi Controller and Service Engine VMs is provided by VCF at the platform layer, not by Avi itself. Organizations must configure Avi SSL/TLS profiles, select appropriate cipher suites, and enable FIPS mode where cryptographic module validation requirements apply to activate these controls.

Control SOV-4

> Operational Sovereignty: Operational sovereignty measures the practical ability of EU actors to run, support, and evolve a technology independently of foreign control. It focuses on continuity of operations, skill availability, and resilience against external dependencies. Contributing factors: Ease of migrating workloads or integrating with alternative EU- controlled solutions without vendor lock-in; Capacity for EU operators to manage, maintain, and support the technology without requiring non-EU vendor involvement; Existence of an EU-based talent pool with the expertise to operate and sustain the service; Assurance that operational support is delivered from within the EU and subject exclusively to EU legal frameworks; Availability of full technical documentation, source code, and operational know-how enabling long-term autonomy; Location and legal control of critical suppliers or subcontractors involved in service delivery.

SCF Controls: AAT-20.2, BCD-01, CLD-07, CLD-08, OPS-01.1, PRI-06.6, TDA-20.3

Aggregate Coverage: Contributes

VCF (Contributes)

VCF provides a broad set of infrastructure-level capabilities that support business continuity and disaster recovery planning. While VCF does not itself constitute a Business Continuity Management System, it supplies the technical mechanisms that underpin resilient deployments and contingency operations.

vSAN stretched clusters allow organizations to extend a single vSAN datastore across two geographically separated sites with an independent witness host. When configured with site mirroring, a stretched cluster can tolerate the complete loss of one site while maintaining data availability. vSphere HA, when enabled on a stretched cluster, automatically restarts virtual machines on the surviving site during a site failure. To support this, vSphere HA admission control should be configured to reserve 50% of the cluster's aggregate CPU and memory resources for failover, and site-specific isolation response addresses should be defined so that vSphere HA can validate host isolation even when network connectivity between sites is lost. The admission control setting for host failures the cluster tolerates should also be updated after cluster expansion to reflect the correct host count per availability zone. When stretching a workload domain cluster, the default management vSphere cluster must already be stretched. vSphere HA uses network heartbeats to validate the state of VMware ESX hosts, and can also use heartbeat datastores mounted to more than one host (excluding vSAN datastores) for cluster membership monitoring.

vSphere HA also operates at the individual cluster level independent of stretched clusters. The HA configuration remediation mode can be set to quarantine mode, maintenance mode, or mixed mode to handle host failures in a controlled manner, helping maintain workload availability during hardware events. vSphere Fault Tolerance provides a higher tier of availability for individual virtual machines by maintaining a secondary VM in continuous lockstep with the primary, enabling instantaneous failover with no data loss if the primary host fails.

vSAN Data Protection provides snapshot-based protection through the vSAN Snapshot Service. Protection groups support configurable snapshot scheduling with adjustable intervals and retention policies that define retention duration and time units. This allows organizations to define data protection policies aligned with their recovery point objectives. The Snapshot Service API also supports operations such as restoring deleted virtual machines from snapshots and creating linked clones from snapshots.

vSphere Replication provides asynchronous replication of virtual machines with a configurable Recovery Point Objective (RPO), allowing organizations to replicate workloads to a secondary site and define acceptable data loss windows as part of their contingency planning.

VCF requires all management components, including VCF Operations, VMware vCenter, NSX, and VCF Automation, to be configured with a regular backup schedule and retention policy. VCF Operations configures NSX Manager backups during the initial deployment, with a recommended backup frequency of hourly and 7-day retention. Backup jobs for VCF Operations and all vCenter instances should start within the same 5-minute window to capture a consistent state. The VMware Infrastructure Health adapter within VCF Operations monitors backup job retention for management components, including NSX Manager. Changed Block Tracking (CBT) tracks modified disk sectors on virtual machines to support incremental backups, reducing backup windows and storage requirements for VM-level backup operations. vCenter itself also supports a

High Availability configuration with Active and Passive nodes whose configuration files are continuously replicated, providing management-plane resilience independent of the backup schedule.

VMware HCX provides workload mobility capabilities for migrating and replicating VMs between sites. HCX supports multiple migration types through its Mobility Agent service, including vMotion, Cold Migration, Bulk Migration, and Replication Assisted vMotion (RAV), each addressing different scenarios from large-scale parallel migrations to zero-downtime live migrations. HCX also includes appliance-level backup and restore capabilities for the HCX Manager, including certificate backup and restore, supporting planned migrations and recovery scenarios as part of broader BC/DR operations.

VCF Operations monitors infrastructure health and surfaces alerts when conditions threaten availability, giving operations teams early warning of potential failures. The Platform Health Alerts page provides a centralized view of alerts across data sources and infrastructure nodes. VCF Automation provides alerts through the associated VCF Operations application for capacity, performance, and availability monitoring. VCF Operations also includes infrastructure planning and what-if analysis capabilities that help organizations forecast capacity requirements and plan for contingency scenarios. Committed scenarios reserve capacity by setting aside resources for new, upcoming, or planned workloads, helping stakeholders understand capacity requirements across the environment. Integration with IT Service Management and Configuration Management Database platforms is supported through VCF Automation, enabling organizations to tie their infrastructure management into broader service continuity workflows.

Lifecycle management tools allow administrators to plan updates and schedule maintenance windows to make changes with minimal disruption to services, supporting the operational continuity aspect of business continuity planning.

While VCF provides these technical capabilities, a complete BCMS requires organizational elements that fall outside the platform: documented continuity of operations plans, recovery time and recovery point objectives defined by business stakeholders, regular testing and exercise of recovery procedures, communication plans, and management oversight of the continuity program. VCF supplies the infrastructure resilience and recovery mechanisms that such plans depend on, but the planning, governance, and procedural components must be established and maintained by the organization.

VMware vDefend (Contributes)

VMware vDefend contributes to business continuity by providing security-layer continuity mechanisms that remain active during disaster recovery events and workload migrations. NSX Federation Security maintains security posture across data centers during disaster recovery and workload migration, preserving distributed firewall and gateway firewall policies so that enforcement continues through failover.

The Security Services Platform, which provides the underlying infrastructure for vDefend's threat detection and inspection services, supports vSphere High Availability configurations. When a VMware ESX host becomes unhealthy, vSphere HA migrates the affected Security Services Platform virtual machines to healthy hosts in the cluster, restoring threat detection capacity without manual intervention. VMware recommends enabling vSphere HA for host-level failure recovery and configuring a storage policy with RAID for datastore-level failure tolerance in Security Services Platform deployments.

Security Services Platform can also be deployed on a vSAN stretched cluster configuration, which replicates storage across two physically distinct sites connected by a high-bandwidth or low-latency inter-site link. This configuration requires a minimum of two sites with their own hosts and a witness host to handle potential conflicts arising from network partitions, supporting site-level failure scenarios.

When vSphere HA restarts workload VMs on a new recovery host after a failover, the vDefend distributed firewall automatically reinspects all traffic from those VMs on the receiving host, because the Flow Table contents cannot be moved between hosts during host recovery. Security policy enforcement resumes on the new host without manual reconfiguration.

The broader BCMS framework, including Continuity of Operations Plan and BC/DR playbooks, impact analyses, and organizational governance processes, is outside vDefend's scope and must be established through dedicated business continuity management programs.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) provides the technical execution layer for contingency planning in virtualized infrastructure. Recovery plans define the sequence, priority, and configuration for failover operations, acting as executable BC/DR playbooks. Protection groups organize virtual machines into logical units aligned with application tiers or business functions, establishing the scope of each contingency plan.

Recovery plans support non-disruptive testing before use in planned migration or disaster recovery, allowing organizations to validate their contingency procedures without affecting production workloads. Recovery plan configuration differs between emergency scenarios at the protected site and planned migration of services from one site to another, with PNR supporting both modes. A forced recovery function is available during disaster recovery plan execution for scenarios requiring immediate failover.

vSphere Replication provides host-based replication with configurable Recovery Point Objectives (RPO), allowing organizations to set per-workload data loss tolerances. vSphere Replication supports up to 200 point-in-time recovery instances per virtual machine using the Latest Instances option, enabling recovery to specific earlier states. This capability is relevant to contingency scenarios involving data corruption or ransomware events. When the source site is accessible at the time of recovery, vSphere Replication can synchronize recent changes from the source site to the target site before completing the operation, reducing data loss.

In 9.1, PNR extends to multi-tenant disaster recovery through integration with VCF Automation. The product implements role-based access control with Organization Administrator, Project Administrator, and Project Advanced User roles, each with defined authorization levels for disaster recovery operations. Role assignments must be configured on both the primary and secondary PNR sites to maintain consistent access control across the DR environment. A multi-region deployment topology places separate management domains in each region to contain disaster impact to one region, while enabling service recovery in an alternate management domain when VCF Automation and VCF Operations are affected. The VCF Operations Orchestrator Plug-In for PNR provides Configuration, Inventory Mappings, Protection Groups, and Storage predefined workflow categories for workflow-driven recovery operations.

Cyber recovery deployments support a standard VCF topology where the workload domain is isolated from the management domain, providing separation between management and user workloads in clean-room recovery scenarios.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) supports the use of secure protocols across its communication channels, contributing to organizational requirements for secure data handling in cloud-hosted database services.

For client-to-database connectivity, DSM database clusters use TLS certificates for both client connections and monitoring traffic. Clients connecting to a DSM-managed database cluster must add the cluster's certificate authority (CA) to the operating system or browser trust store to establish a TLS-encrypted session. The DSM UI and API both provide mechanisms to retrieve the CA for this purpose. DSM supports custom trusted certificate configuration; custom certificates must comply with FIPS requirements. DSM service components also use trusted certificates when connecting to external TLS-secured systems.

For connections to the Provider VM, DSM requires that the Provider VM's CA be added to the list of trusted CAs to establish a secure TLS connection. The CA must be stored in the trust store and used to validate the Provider VM certificate.

DSM applies secure protocol requirements across all supported database engines. PostgreSQL replication connections support SSL/TLS encryption with a configurable SSL mode (`sslmode` parameter) to secure data in transit between source and target databases. For ongoing client access, PostgreSQL databases support SSL-encrypted connections with downloadable CA certificates. MySQL replication operations include SSL trust validation: DSM verifies network access to the source database and validates SSL trust before proceeding. SQL Server clusters use TLS to encrypt communication between clients and the database cluster.

The Consumption Operator requires a TLS certificate for secure communication between the operator and the DSM provider. A Cloud Administrator must obtain and save this certificate when enabling self-service consumption of data services.

For outbound management communications, DSM supports Auto TLS configuration for SMTP server connections, available under Managing Log Forwarding Settings in the administration interface.

Access to data management operations is controlled through DSM's role-based access control model. Administrators have full management and monitoring capabilities across all data services, while users are scoped to data services they are authorized to access. This scoping restricts who can initiate import, export, or configuration operations within the platform.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer provides application-layer resilience mechanisms at both the control plane and the data plane that organizations can incorporate into business continuity plans for application delivery services.

At the control plane, the Avi Controller is deployed as a cluster with continuous configuration replication across all nodes, providing management-plane redundancy so that a single Controller node failure does not interrupt application delivery operations. Avi supports deployment of the Controller cluster across two availability zones, with vSphere HA recommended to support recovery of any node that fails in the secondary zone. For single-node deployments, enabling vSphere HA on the Controller VM is recommended for node recovery following a host failure. The Avi Controller provides configuration backup and restore capability that supports disaster recovery and configuration migration across Controller instances; organizations should capture and retain Controller backups as part of their continuity documentation.

Availability Zone awareness in Avi allows multiple availability zones to be configured, each defining a subset of vCenter instances, hosts, and datastores that share a fault domain. Configuring availability zones during Service Engine Group setup distributes data-plane capacity across isolated failure domains, supporting continuity designs where application delivery capacity remains available if one fault domain becomes unavailable.

At the data plane, Avi's Elastic High Availability mode provides fast recovery for individual virtual services following Service Engine failure. In vSphere DRS-enabled clusters, virtual service recovery occurs before vSphere HA brings up a replacement Service Engine when virtual service slots are available in other Service Engines, giving the application delivery layer its own recovery path distinct from the infrastructure restart cycle. If vSphere HA does not recover the failed Service Engine within a configured timeout period, the Avi Controller resumes recovery independently. Service Engine VMs should be deployed on vSphere clusters configured with both HA and DRS in fully automated mode; when Avi is deployed with VMware Kubernetes Service (VKS) on VCF, DRS and HA are required on every vSphere cluster used with Avi.

For Kubernetes environments, the Avi Multi-Cluster Kubernetes Operator (AMKO) provides federation architecture that supports disaster recovery across Kubernetes clusters. AMKO designates a leader cluster and multiple member clusters, coordinating application delivery failover as part of a multi-cluster DR strategy.

Avi ControlScripts, used in conjunction with the alert framework, can perform automated failover actions such as adding or removing routes based on VIP attachment and detachment events, allowing organizations to define automated remediation workflows that respond to system events without manual administrator intervention.

In 9.1 deployments using VCF Automation, Avi's Central Licensing maintains a global capacity pool that removes the requirement for duplicate licenses in DR scenarios, supporting license continuity across failover sites.

Organizations should document these mechanisms, including the Elastic HA configuration, Controller cluster topology and backup schedule, availability zone placement, and AMKO federation design, in their Business Continuity and Disaster Recovery playbooks. The technical recovery capabilities are built into the product; the planning artifacts, testing procedures, and escalation workflows are organizational responsibilities that extend beyond what Avi provides.

Control SOV-5

> Supply Chain Sovereignty: Supply chain sovereignty evaluates the geographic origin, transparency, and resilience of the technology supply chain, focusing on the extent to which critical components and processes remain under EU control or exposed to non-EU dependencies. Contributing factors: Geographic source of key physical parts, manufacturing location - countries where hardware is manufactured or assembled; Jurisdiction and provenance of embedded code controlling hardware, firmware; Origin of Software: where and by whom software is architected and programmed, location and jurisdiction governing software packaging, distribution, and updates; Degree of reliance on non-EU vendors, facilities, or proprietary technologies; Visibility into the entire supplier and sub-supplier chain, including audit rights.

SCF Controls: AST-03.2, AST-13, MON-16.2, RSK-09, RSK-09.1, TDA-04.2, TPM-01, TPM-03

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control SOV-6

> Technology Sovereignty: Technology sovereignty evaluates the degree of openness, transparency, and independence in the underlying technological stack, ensuring EU actors can interoperate, audit, and evolve solutions without lock-in to foreign proprietary systems. Contributing factors: Ability to integrate with other technologies through well-documented and non-proprietary APIs or protocols, extent to which the solution adheres to publicly governed and widely adopted standards, reducing dependency on single vendors; Whether software is accessible under open licenses, with rights to audit, modify, and redistribute, ensuring transparency and adaptability; Visibility into the design and functioning of the service, including architectural documentation, data flows, and dependencies; Degree of EU independence in high-performance computing capabilities, including processors, accelerators, and software ecosystems.

SCF Controls: AAT-20.2, CFG-04.1, CLD-07, CLD-08, TDA-04.2

Aggregate Coverage: Contributes

VCF (Contributes)

VCF contributes to AI implementation documentation by providing documented compute and hardware resource requirements, event log collection mechanisms, and infrastructure configuration references that support the documentation of AI deployment environments. VCF's compatibility guides document supported hardware, VCF Operations tracks infrastructure configuration, and VCF Log Management documents the event collection architecture.

Documenting AI model architecture, training methodology, and decision logic is outside VCF's scope.

VMware Private AI Services (PAIS), formerly Private AI Services, provides documentation and platform features that address several of the listed implementation documentation requirements. For computational and hardware resource requirements (item 6), the platform documents the system architecture of PAIS deployments, including GPU-accelerated workload domain requirements, vGPU profile configurations, VM class definitions for AI workloads, and the NVIDIA GPU and driver compatibility requirements. VI administrators configure vGPU-based VM classes that define the GPU resource allocations available to AI workloads. The PAIS catalog items in VCF Automation, including AI Workstation, DSM Database, Triton Inferencing Server, and AI Kubernetes Cluster, document the components users can provision in selected namespaces, and the deployment Overview tab summarizes the installed software and the access points for the application, services, and workstation VM. PAIS setup is documented for both connected and disconnected (air-gapped) deployment models, with a Harbor registry in the Supervisor or another OCI-compliant registry used to store validated ML models.

For event logging mechanisms (item 8), the platform documents multiple logging and telemetry capabilities. Deep Learning VMs record cloud-init script logs in `/var/log/dl.log` and GPU driver installation logs in `/var/log/gpu-install.log`, and they provide JupyterLab instances and console logs for verifying component installation and operation. The vLLM inference server logging level is configurable through the `VLLM_LOGGING_LEVEL` environment variable. PAIS Controller pod metrics are exposed at a Prometheus endpoint on the `pais-controller-manager` service and are integrated into observability platforms through Telegraf configuration in the Supervisor instance, with VCF Operations as the documented consuming platform. Deep Learning VMs also expose vGPU performance metrics through Prometheus. PAIS supports LLM traces collection in an OpenTelemetry Collector by configuring the `spec.observability.llmTraces` section of the `PAISConfiguration` custom resource. Knowledge Base operations track crawled documents, new documents found, deleted documents, modified documents requiring embeddings updates, and skipped documents, and record timing metrics for download, embedding generation, and the overall index operation.

For human oversight measures (item 5) and for interpreting outputs (items 4 and 5), MLOps engineers analyze LLM traces and Prometheus metrics to inspect agent execution flow across components and knowledge base indexing. The PAIS Agent Builder maintains a list of approved tools for use in AI agents and requires examination and approval of MCP server tools and capabilities before they are used by agents. Knowledge Bases are created with linked data sources such as Microsoft SharePoint sites and Amazon S3-compatible stores, which gives implementers a documented data lineage for retrieval-augmented generation outputs. Access to PAIS is controlled through OIDC authorized groups, where user accounts must be members of specified groups to consume the service, and personal access tokens generated from user account settings provide programmatic access to Knowledge Bases and data sources.

Requirements 1 (provider contact details), 2 (characteristics, capabilities, and limitations of the AI system), 3 (errata from initial conformity assessment), and 7 (projected useable lifetime) are documentation obligations carried by model vendors and the deploying organization rather than by the infrastructure platform. PAIS supplies registry and catalog mechanisms that organizations can attach revision and provenance metadata to, but the substantive documentation content for those items is outside the platform's scope.

VMware vDefend (Contributes)

VMware vDefend contributes to secure protocol enforcement for the import, export, and management of data through certificate management, protocol governance, and communications security within the Security Services Platform.

The Security Services Platform uses certificates and secure communications protocols to protect transmissions between vDefend product components. When establishing external communications, the platform performs TLS handshake negotiation and actively rejects configurations that use deprecated algorithms, insufficient key lengths, or older protocol versions, applying this enforcement through FIPS-validated cryptographic modules.

The Security Services Platform's Security Segmentation capability identifies risky application protocols that transport data in clear text, and provides guidance for administrators on enforcing minimum secure protocol versions across the environment. The vDefend distributed firewall can then enforce network-level policy to restrict or block workload use of insecure protocols, limiting data movement to approved secure communication paths. Security policies applied to VPC environments are translated and enforced on transport nodes, providing consistent protocol controls across distributed infrastructure.

Certificate lifecycle management within the Security Services Platform supports importing certificates signed by customer-managed certificate authorities, enabling organizations to integrate vDefend into their existing PKI infrastructure. Certificate consistency across Security Services Platform components, including backup and restore operations, is required to maintain uninterrupted secure communications for services such as Network Detection and Response sensors.

For integrations with external systems such as SIEM platforms, SSL verification is recommended to remain active for Network Detection and Response SIEM integration to reduce the risk of interception and modification of security telemetry in transit. Administrators can also configure whether Network Detection and Response flow data is shared to the cloud for advanced threat analysis, providing governance over what security data leaves the on-premises environment.

While vDefend secures its own component communications and provides network-level protocol governance for workload traffic, the broader scope of secure protocols across all cloud infrastructure layers, including storage encryption, management API security, and VM migration, is addressed by other VCF components including VMware vCenter and vSAN.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) uses secure protocols for all data movement and management communications in its disaster recovery and replication operations.

All replication traffic between protected and recovery sites is protected using TLS. Starting with vSphere Replication 9.0.3 and later, network encryption of replication traffic is mandatory for both new and existing replications. For configurations on earlier versions, network encryption is available as a configurable option during replication setup.

PNR uses TLS certificates and private keys to protect network communications and authenticate with other servers. An SSL/TLS certificate is required as the endpoint certificate for all TLS connections established to the PNR server, and custom certificates must meet specific product-defined requirements. The PNR Appliance supports importing CA-signed server certificates in .pem format through its management interface. Administrators can configure the minimum TLS protocol version through the envoy-proxy-config.json file.

All communications between PNR and VMware vCenter instances take place over TLS connections, covering control plane traffic used to configure, monitor, and manage replication pairs, protection groups, and recovery plans. The VCF Operations Management Pack for PNR uses HTTPS on port 443 when connecting to the PNR appliance. The ransomware recovery component uses TLS for secure communications with connected systems.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) supports the use of secure protocols across its communication channels, contributing to organizational requirements for secure data handling in cloud-hosted database services.

For client-to-database connectivity, DSM database clusters use TLS certificates for both client connections and monitoring traffic. Clients connecting to a DSM-managed database cluster must add the cluster's certificate authority (CA) to the operating system or browser trust store to establish a TLS-encrypted session. The DSM UI and API both provide mechanisms to retrieve the CA for this purpose. DSM supports custom trusted certificate configuration; custom certificates must comply with FIPS requirements. DSM service components also use trusted certificates when connecting to external TLS-secured systems.

For connections to the Provider VM, DSM requires that the Provider VM's CA be added to the list of trusted CAs to establish a secure TLS connection. The CA must be stored in the trust store and used to validate the Provider VM certificate.

DSM applies secure protocol requirements across all supported database engines. PostgreSQL replication connections support SSL/TLS encryption with a configurable SSL mode (`sslmode` parameter) to secure data in transit between source and target databases. For ongoing client access, PostgreSQL databases support SSL-encrypted connections with downloadable CA certificates. MySQL replication operations include SSL trust validation: DSM verifies network access to the source database and validates SSL trust before proceeding. SQL Server clusters use TLS to encrypt communication between clients and the database cluster.

The Consumption Operator requires a TLS certificate for secure communication between the operator and the DSM provider. A Cloud Administrator must obtain and save this certificate when enabling self-service consumption of data services.

For outbound management communications, DSM supports Auto TLS configuration for SMTP server connections, available under Managing Log Forwarding Settings in the administration interface.

Access to data management operations is controlled through DSM's role-based access control model. Administrators have full management and monitoring capabilities across all data services, while users are scoped to data services they are authorized to access. This scoping restricts who can initiate import, export, or configuration operations within the platform.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer provides SSL/TLS termination and re-encryption capabilities that address secure protocol requirements for application-layer data in transit. Each virtual service that handles secure connections requires an SSL/TLS profile, which administrators configure to specify the cipher suites and protocol versions the service will accept. This makes protocol enforcement an explicit, per-service configuration that must be defined when deploying application services on Avi.

Avi operates as a proxy: client-facing connections terminate at the Service Engine, and the Service Engine re-encrypts traffic to backend application servers using a separate SSL/TLS configuration. When Service Engine-to-server re-encryption is enabled, the Service Engine can present a client certificate to backend servers, supporting mutual authentication on the downstream connection. This dual-termination model supports organizations in applying protocol requirements at both the client-facing and server-facing legs of an application connection, and in enforcing the requirement that all forwarded requests remain encrypted end-to-end.

In VCF-integrated deployments, the Avi cloud connector supports a `verify_certificates` option that uses the PKI profile with the System Trust Store to validate certificates when establishing connections to NSX Manager and vCenter. The Truststore PKI Profile flag must be set on the cloud configuration to enable certificate validation for cloud connectors. This extends secure protocol requirements to the control-plane connections between Avi and its VCF infrastructure dependencies, not only to application traffic.

For Kubernetes workloads managed through the Avi Kubernetes Operator (AKO), PKIProfile can be configured on ingress routes to enable certificate-based authentication and encryption for backend service communication. This allows the same SSL/TLS enforcement model to apply to cloud-native application traffic routed through Avi.

Control SOV-7

> Security & Compliance Sovereignty: Security & Compliance sovereignty measures the extent to which security operations, compliance obligations, and resilience measures are controlled within the EU, ensuring independence from foreign jurisdictions and long-term operational assurance. Contributing factors: Attainment of EU and internationally recognized certifications (e.g., ISO, ENISA schemes); Adherence to GDPR, NIS2, DORA, and other EU frameworks; Security Operations Centres and response teams operating exclusively under EU jurisdiction, control over security monitoring/logging - customer or EU authority ability to oversee logs, alerts, and monitoring functions directly; Transparent, timely, and EU-compliant reporting of breaches or vulnerabilities, maintenance Autonomy - ability to develop, test, and apply security patches independently of non-EU vendors; Capacity for EU entities to perform independent security and compliance audits with full access.

SCF Controls: CPL-01, CPL-03, CPL-03.1, IRO-10, IRO-10.2, MON-01, MON-06, VPM-02, VPM-05

Aggregate Coverage: Contributes

VCF (Contributes)

VCF provides enterprise-wide monitoring through integrated tools that span infrastructure health, performance metrics, compliance assessment, log analysis, and alerting.

VCF Operations continuously monitors the operational state of all VCF components to detect conditions that may affect availability or compliance. It collects metrics across compute, storage, and networking resources, and correlates them over time to identify trends that may lead to potential failures. VCF Health, a component of VCF Operations, continuously monitors the operational state of VCF components to verify they are functioning as expected, giving platform operators a single view of operational status. Platform operators can configure alerts, perform log analysis, run infrastructure diagnostics, and review VCF health from a unified interface.

VCF Operations provides continuous compliance management by monitoring infrastructure against industry-standard benchmarks and custom policies. Administrators can create custom compliance benchmarks that monitor selected alert definitions and policies, tailoring the monitoring posture to organizational requirements. The platform assesses objects based on selected compliance policies and activates alert definitions associated with the current scorecard, providing ongoing visibility into compliance status. Compliance-related alerts are triggered on objects and displayed through a dedicated compliance alert subtype.

The alerting subsystem in VCF Operations supports the full lifecycle of alert management. Administrators can configure and manage alert definitions to monitor infrastructure events and conditions, including performance metrics, log events, and custom conditions configured through the Management Pack Builder. The Platform Health Alerts page provides a centralized view of all alerts related to overall system health, including alerts from data sources and infrastructure nodes. Health alerts are generated for conditions that affect the health of the environment and require immediate attention. Administrators can create policies to govern how alert definitions are evaluated, including which objects to monitor and what thresholds to apply, and can apply consistent practices to optimize alert behavior across monitored objects.

VCF Operations supports flexible grouping and scheduling to align monitoring with operational realities. Custom groups can be configured with automatic membership criteria to support monitoring strategies and achieve a high level of automation. Maintenance schedules can be assigned to objects undergoing planned downtime to keep monitoring data accurate and avoid false alerts when an object is offline for scheduled maintenance.

VCF Log Management provides centralized log collection and analysis. It ingests logs from VMware vCenter, VMware ESX hosts, NSX, vSAN, VCF Automation, VCF Operations, VCF Identity Broker, VCF Operations for Networks, and VCF Operations HCX. VCF Operations and VCF Log Management must be integrated to implement the centralized log collection architecture. The platform provides dashboards and analytics for troubleshooting and security investigation. Log-based alert definitions allow administrators to trigger notifications when specific log patterns appear across monitored systems. Log forwarding to external SIEM or syslog targets is configurable, with support for TLS encryption to protect log data in transit.

The vCenter syslog forwarding API supports TCP, UDP, TLS, and RELP protocols. NSX Manager syslog supports TLS and requires port 6514 for encrypted log transmission.

At the vSphere layer, the alarm infrastructure integrates with events and performance counters to provide real-time alerting. The `MetricAlarmExpression` data object allows administrators to set alarms on specific performance metrics with configurable monitoring intervals and aggregation functions for processing data points. The vSphere Client provides direct visibility into tasks and events for change tracking and troubleshooting. VCF ships with preconfigured vSphere alarms that are active by default and monitor specific components including the Identity Management Service, Message Bus Configuration Service, ESX Agent Manager, VMware Service Control Agent, vSphere HA host status, and host connection and power state. The vCenter Management Interface provides appliance-level health visibility through an Overall Health badge that reflects one of five states (Good, Warning, Alert, Critical, or Unknown) across all vCenter components, and notifies administrators of available security patches through Critical or Alert badge indicators when automatic patch checks are enabled.

When VCF Automation is integrated with VCF Operations, the Insights dashboard and Alerts tab provide real-time capacity awareness and alert information for VCF Automation objects, giving IT teams visibility into resource usage across both provisioned workloads and the underlying infrastructure. The VCF Automation Monitor tab surfaces deployment health information sourced from VCF Operations, and deployment cards display Last Request Status information that tracks the most recent operation on each deployment (Create, Update, or Delete) and its outcome (Successful, In Progress, or Failed). The Management Pack Builder allows administrators to set up event definitions for integrated systems, monitoring specific conditions or thresholds and configuring alerts for proactive management. Management Packs extend monitoring to network devices, including sensor monitoring for devices such as switches and routers. Adapters can be installed on VCF Operations nodes to extend monitoring to additional systems and data sources beyond the core VCF components.

The VMware Kubernetes Service (VKS) layer of VCF extends enterprise monitoring to container workloads and Kubernetes infrastructure. VKS clusters support `MachineHealthCheck` objects for both control plane nodes and worker node deployments; administrators manage these through the VCF CLI `vcf cluster machinehealthcheck` command, which supports configuration of unhealthy-condition definitions, maximum unhealthy thresholds (expressed as an absolute number or a percentage), and label selectors to target specific machines. The VCF CLI `vcf cluster machinehealthcheck node get` and `vcf cluster machinehealthcheck control-plane get` commands retrieve node and control plane health status on demand. VKS cluster health conditions, including the `Ready` and `ControlPlaneReady` status indicators, reflect the operational state of each cluster and are accessible via `kubectl` and the VCF CLI. The VCF Automation Provider Management UI includes a Services overview interface that exposes VKS cluster management functions and the health status of all cluster elements, giving platform operators a centralized view of the VKS tier. The Kubernetes API server within VKS clusters exposes diagnostic and health check endpoints (`/healthz`, `/livez`, `/readyz`, `/statusz`, and `/metrics`) that external monitoring systems can scrape to track API server availability and readiness; the `kube-apiserver` also emits the `kubernetes_healthchecks_total` counter and `kubernetes_healthcheck` gauge metric, which report structured health check status for components including `etcd` and `autoregister-completion`. Access to these monitoring endpoints is governed by the built-in Kubernetes system:monitoring RBAC group, which grants read access to liveness, readiness, and metrics endpoints while restricting broader cluster access. The Kubernetes Watch API allows monitoring tools to track changes to any API object as a live stream, supporting event-driven monitoring without continuous polling. The Node Problem Detector monitors node health and surfaces node-level problems as Kubernetes events and node conditions that external monitoring systems can consume via the event stream.

For workload-level monitoring within VKS clusters, Prometheus Operator is available as a standard package and defines custom resource definitions to deploy and manage Prometheus and Alertmanager `StatefulSets`, enabling high-availability alerting configurations. The Prometheus Alertmanager custom resource definition configures alert grouping, inhibition, and routing to external notification systems. The Istio service mesh, also available as a standard package on VKS clusters, provides service-level monitoring alongside its traffic management capabilities. The Dynatrace Operator can be deployed on VKS clusters to automate the deployment, scaling, and maintenance of the Dynatrace monitoring stack. Historical analysis, dashboarding, and alerting on cluster-level metrics require a dedicated monitoring solution configured to scrape the VKS metrics endpoints and subscribe to the Kubernetes event stream. The `vcf cluster support-bundler create` command gathers diagnostic information from Supervisor and VKS clusters when troubleshooting is needed.

At the network layer, NSX provides monitored indicators including node uptime status, deployment status, management and controller connections, tunnel status, PNIC status, and uptime duration. NSX time series APIs allow viewing data such as VPN statistics, including metadata about the number of tunnels configured and how many are up or down at a given time.

VCF Operations also supports self-monitoring through the VCF Operations adapter, which collects internal metrics such as activity success counts. This allows administrators to verify that the monitoring platform itself is functioning correctly. The VCF Operations policy engine monitors conditions and generates events to react to changing states in the orchestrator server or connected technologies. The Adapter Instance monitoring interval is configurable, with intervals as granular as five minutes.

VMware Private AI Services (PAIS) includes a built-in observability stack that supplies monitoring data for AI infrastructure and AI workloads. PAIS Controller metrics are exposed at a Prometheus metrics endpoint on the pais-controller-manager service, allowing collection by monitoring systems and visualization in observability platforms such as Grafana. Observability is activated by updating the PAISConfiguration custom resource with optional sections for Prometheus metrics collection and LLM trace forwarding. The Prometheus runtime can be configured with a specified storage class policy and a configurable metrics retention period, giving administrators control over how long monitoring data is stored. A Prometheus server pod must be in a ready and running state for observability to function.

LLM trace data from PAIS can be forwarded to an OpenTelemetry Collector endpoint over http/protobuf or gRPC, providing distributed tracing for AI model inference activity. The observability.IlmTraces configuration governs trace forwarding and works alongside the Prometheus metrics pipeline to produce two signals: metrics for health and performance, and traces for inference-level behavior. PAIS Controller metrics include the Model Endpoint Controller and PAIS Configuration Controller, giving monitoring platforms component-level visibility into PAIS operations.

In 9.1, PAIS adds direct integration with VCF Operations. The controller-pod-metrics-streaming setting activates streaming of PAIS controller pod metrics from the Supervisor cluster to VCF Operations, allowing administrators to view PAIS metrics alongside other VCF infrastructure metrics. A Telegraf deployment in the vmware-system-monitoring namespace uses a telegraf-user-config ConfigMap to manage user-configured monitoring endpoints for PAIS. The VCF Consumption CLI connects to the Supervisor endpoint and enables VI administrators to stream metrics from the PAIS controller pod. DevOps engineers can visualize health and performance metrics for PAIS components running in a VCF Automation namespace using Grafana, and MLOps engineers and application developers can monitor the health, quality, and behavior of AI applications and the models used in agents through the same observability platforms.

PAIS supplies the technical monitoring mechanisms this control requires. Enterprise-wide orchestration of monitoring controls, including alert threshold definition, escalation procedures, and integration with a centralized monitoring or log management platform, remain organizational responsibilities.

VMware vDefend (Contributes)

VMware vDefend (vDefend) supports network security monitoring through multiple integrated capabilities that contribute to enterprise-wide monitoring frameworks.

Security Intelligence, hosted on the Security Services Platform (SSP), provides continuous visibility into network traffic flows across protected workloads. It monitors flows between application pairs, updates the Policy Protection Status for each application to indicate that flows are being monitored, and continuously detects new flows between environment pairs, alerting administrators to unauthorized or unexpected traffic. The Monitor & Plan Overview page presents a consolidated view of pending actions, data collection activity status, network security posture, flow trends, and a summary of suspicious traffic. The Security Intelligence Recommendations monitoring capability checks for changes in the scope of monitored entities every hour when the Monitoring toggle is enabled, allowing administrators to keep policy recommendations current as environments evolve. Segmentation Monitoring provides dedicated views of application and infrastructure flows with configurable time ranges spanning from the last hour to the last month, and displays flows related to detected infrastructure servers in a specified time interval.

The Network Detection and Response (NDR) Sensor checks the status of activated detection features every 15 minutes and updates its configuration accordingly, supporting sustained threat detection coverage. NDR Sensor health status monitoring tracks whether key components, including network traffic capture, file analysis, and data transmission, are operating properly.

The NDR Sensor also sends alerts notifying administrators of conditions such as services being down, high CPU, memory, or disk usage, packet drops, or missed heartbeats. NDR provides role-based access control covering triage operations such as viewing detections and campaigns, as well as SIEM configuration management.

The Security Services Platform monitors its own health, with Yellow or Red health status indicators displayed when issues occur in subcomponents. SSP monitoring tracks the status of the flow clustering job that runs every hour and reports suspicious traffic detector execution status after each run, with values such as SUCCESS, NOT_ENOUGH_BASELINE, and FAILURE.

The Security dashboard provides a consolidated view for monitoring and configuring security features across network workloads. In federated deployments, the Global Manager provides centralized event management and monitoring for malicious IP blocking activity across all Local Managers.

vDefend generates Unified Security Logs that VCF Log Management can collect and analyze. These logs cover Distributed Firewall, Gateway Firewall, Identity Firewall, FQDN Analysis, TLS Inspection, Intrusion Detection/Prevention System (IDPS), and URL Filtering activity. Broader log aggregation and SIEM correlation for enterprise-wide monitoring remain the responsibility of the SIEM and logging infrastructure deployed alongside vDefend.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) provides monitoring capabilities specific to the disaster recovery domain and integrates with the broader VCF monitoring ecosystem through dedicated Protection and Recovery dashboards in VCF Operations. VCF Operations collects configuration, component health, and usage information from protection and recovery servers and site pairs, presenting this through an overview dashboard that displays all site pairs, configurations, health scores, and protection and recoverability status for sites and virtual machines. PNR provides default health alerts for protection groups, recovery plans, cloud file systems, recovery SDDCs, and protected sites through VCF Operations, and the dashboards surface alerts informing administrators about configurations and deployments that require attention. Separate dashboard views cover vSAN Snapshots and Replication monitoring, and an Inventory view provides health status, metrics, alerts, and network topology for all monitored objects.

Within its own domain, PNR provides event logging and configurable alarms for replication and recovery operations. Administrators can create alarms that trigger on specified event conditions, with support for email notifications, SNMP traps, or script execution in response. PNR monitors connections between sites by sending periodic ping requests, logs connection check events, and generates alarms when broken connections are detected. The alarm system generates warning-level alerts for expired snapshot schedules, replication errors such as missing VMs, and connectivity issues with protected and target sites. PNR also monitors resource consumption on the Protection and Recovery Server host and raises alarms when configurable resource thresholds are reached.

PNR includes a health monitoring service that checks each health finding every five minutes and records a timestamp at each check. Health findings are tracked with status history that can be reviewed over configurable time periods, defaulting to a 24-hour window. The Protection and Recovery Appliance supports syslog forwarding to a remote syslog server for log analysis and centralized monitoring, allowing appliance-level events to flow into existing log management infrastructure.

The Clean Room Orchestrator, part of PNR's Cyber Recovery capabilities, supports monitoring of clean room events, tasks, and alarms for recovery operations. Clean Room Orchestrator can forward events to VCF Operations, Microsoft Sentinel, or Splunk Cloud for centralized monitoring, supporting organizations that require DR event data to flow into enterprise SIEM and log management platforms as part of a broader monitoring strategy. The Clean Room Orchestrator integrates with endpoint detection and response tools, including CrowdStrike and Carbon Black sensor appliances, for threat detection within isolated recovery environments.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) contributes to enterprise-wide monitoring controls through a layered set of observability capabilities covering database health, infrastructure compliance, and event-driven alerting.

DSM provides a built-in monitoring framework that tracks database instance health across multiple dimensions. From the Monitoring tab in the DSM UI or the vSphere Client, administrators and authorized users can view real-time and historical metrics for database VMs, including CPU Health, Data Disk Health, System Disk Health, Max Connections, Active Connections, Write Throughput, Read Throughput, Thread Resource Utilization, Slow Queries per Second, InnoDB Buffer Usage, and InnoDB Reads and Writes. The Monitoring page defaults to the last three hours of aggregated metric data, with the time range adjustable through the Time Range Last dropdown. This visibility supports detection of resource contention, performance degradation, and connectivity anomalies at the database layer.

DSM's alert framework provides configurable thresholds for warning and critical conditions on key health indicators. Administrators and users can set separate warning and critical thresholds for CPU Health, Data Disk Health, System Disk Health, and Max Connections parameters. When thresholds are exceeded, DSM generates typed alerts such as Database Engine Health, Custom Config Health, and Backup Storage Health alerts. DSM monitors database replication status and raises an alert when a secondary database is not replicating data from the primary. DSM also raises an alert when metrics are no longer being received from a cluster, supporting detection of monitoring gaps. Alerts can be activated or deactivated per data service through the "Enable Alerts for DB Alert" setting, providing granular control over which databases are subject to active alerting. For each alert, DSM provides mitigation suggestions to help administrators resolve issues without disrupting database services.

In addition to database-level alerts, DSM generates global infrastructure alerts that surface vCenter integration and infrastructure policy issues. These include `INFRASTRUCTURE_POLICY_VC_ALARMS`, raised when vCenter resource alarms are active; `INFRASTRUCTURE_POLICY_CLUSTER_COMPLIANT` and `INVALID_SUPERVISOR_INFRASTRUCTURE_POLICY_STATUS`, which reflect cluster compliance state; `VCENTER_CONNECTIVITY`, raised when vCenter connectivity fails and triggering troubleshooting steps that include verifying connectivity and checking authentication logs; and `LicensingNotUpdated` and `vCenterBinding` API status alerts that surface license and API integration conditions. A Global Alerts page in the DSM UI provides a centralized view of these alerts during and after installation and ongoing operation.

In 9.1, DSM integrates with VCF Operations to extend observability beyond the DSM UI. When configured, VCF Operations automatically discovers and monitors PostgreSQL and MySQL clusters in the Ready state and displays a Data Services View dashboard showing active alerts and inventory items. VCF Operations presents performance data across three levels: cluster-level for overall health, node-level for infrastructure resource use, and database-level for KPIs including Active Connections, Commits, Rollbacks, and Deadlocks. This integration requires configuring the connection between DSM and VCF Operations before use.

DSM also adds SQL Server as a supported database type, with dedicated monitoring through an integrated InfluxDB instance and Telegraf agents deployed across the provider appliance and SQL Server workload clusters. The monitoring framework supports metrics-based and status-condition-based alerts for SQL Server, covering database engine degradation, non-operational status, and automated or transaction log backup failure conditions.

DSM supports event notification to external systems through a webhook integration. Webhooks transmit alert notifications containing the triggered date, triggered time, triggered user email, task status, previous status, and task type. This allows DSM alerts to feed into centralized monitoring or incident management platforms maintained by the organization. DSM generates plug-in registration logs that administrators can review for vCenter connectivity errors, and DSM's troubleshooting guidance directs administrators to review active vCenter infrastructure alarms and resolve underlying issues identified through the alert system.

Enterprise-wide monitoring requires organizational decisions about monitoring architecture, log aggregation platforms, alerting policies, and operational response procedures that lie outside DSM's scope. DSM provides the data and integration points; the organization must determine how those signals are collected, correlated, and acted upon across the broader environment.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer (Avi) contributes to enterprise-wide monitoring through application-delivery-layer mechanisms built into the Avi Controller and Service Engines.

The Avi Controller monitors Service Engine metrics and sends alarms and notifications to administrators when resource utilization data exceeds defined thresholds. Administrators configure custom thresholds on Service Engine groups through the Resource Events tab in the Avi UI. The Alert feature processes system events and metrics against user-defined rules and generates alerts that trigger configured actions, including sending an email, a syslog message, or an SNMP trap. Alert notifications can be forwarded to remote syslog servers for integration with centralized log management and monitoring platforms.

ControlScripts extend this event-response capability: a ControlScript can alter Avi configuration or communicate to external systems, including through webhook integrations with external monitoring tools, in response to defined system events. This supports dynamic responses including resource scaling or automated notifications based on health scores from monitoring data.

The Avi Controller and Service Engines track health score changes and SSL certificate expiration events. Avi integrates with external monitoring tools such as Prometheus for metrics collection and analysis. User-Defined Metrics allow administrators to define and extract customized telemetry data for specific compliance or reporting needs through the API. Real-time metrics are available via API but deactivate by default after 30 minutes to conserve resources and must be reactivated when sustained real-time visibility is required.

At the data plane, Avi health monitors track the status of backend servers and Service Engines continuously. Service Engine health monitoring records detailed server operational status, including health monitor name, type, last transition timestamps, rise count, fall count, total checks, and total failed checks. Each health monitor can be assigned its own threshold and timeout period for server health validation. Monitors support HTTP, HTTPS, TCP, and UDP health-check protocols with configurable success and failure thresholds, send intervals, and receive timeouts.

In Kubernetes environments managed by the Avi Kubernetes Operator (AKO), the HealthMonitor custom resource definition (CRD) allows users to define custom health monitoring configurations for backend services with fine-grained control over health check parameters. The Avi Multi-Cluster Kubernetes Operator (AMKO) GlobalDeploymentPolicy supports health monitor references for consistent monitoring across multi-cluster deployments. Within VCF Automation, a HealthMonitor custom resource can create monitors for tracking backend pool member status using TCP, PING, and HTTP check types.

Control SOV-8

> Environmental Sustainability: Environmental sustainability assesses autonomy and resilience of cloud services over the long term in relation to energy usage, dependency and raw material scarcity. Contributing factors: Adoption of energy-efficient infrastructure (e.g., low PUE) and measurable improvement targets; Circular economy practices ensuring reuse, refurbishment, and responsible end-of-life treatment of hardware; Transparent measurement and disclosure of carbon emissions, water usage, and other sustainability indicators; Sourcing of renewable or low-carbon energy to power infrastructure and operations.

SCF Controls: AAT-17.2, CAP-04

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides PAIS-native mechanisms that help organizations assess the environmental footprint of AI workloads. Metrics collection for PAIS is activated by uncommenting the `spec.observability.prometheusRuntime` section of the `PAISConfiguration` custom resource, after which DevOps and MLOps engineers can visualize PAIS health and performance metrics for components running in a VCF Automation namespace through observability platforms such as Grafana. The same observability pipeline lets MLOps engineers and application developers monitor models used in agents, giving operators visibility into the operational characteristics of individual AI workloads across deep learning VMs, VKS clusters, and RAG applications. This telemetry is the raw operational data that organizations use to quantify the energy and resource footprint of AI activity.

PAIS also supports resource efficiency through deployment leases. A lease controls how long a PAIS deployment can exist, so that AI workloads provisioned from VCF Automation catalog items are reclaimed at the end of a defined period rather than continuing to consume GPU and compute resources indefinitely. Disconnected (air-gapped) and connected deployment modes both rely on the same observability and lease controls, so the same sustainability-relevant data and lifecycle controls apply regardless of connectivity model.

Broader sustainability and green-metric monitoring at the cluster and host level is a VCF platform capability and is covered under VCF Coverage. The organizational activity of setting sustainability targets, analyzing collected metrics, and documenting environmental impact is a process obligation outside PAIS itself, which is why the rating is Contributes rather than Meets.

VMware vDefend (Contributes)

VMware vDefend provides automated monitoring of its own security infrastructure components through several complementary mechanisms. The Security Services Platform (SSP) Metrics service gathers performance data visualized across the Platform dashboard. The Metrics API supports time-series queries with filtering by metric names, object IDs, resource type, and granularity, and collects metrics including CPU load, network throughput, edge node datapath CPU utilization (`edge_dp_cpu.avg_percentage_busy_cycles`), and system resource counters. Flow collection provides additional visibility into system performance and resource utilization. The metrics service supplies data for Top N analysis, lifetime counters, and historical trend statistics to support firewall operational troubleshooting. For IDS/IPS operational health, the `get ids engine global stats` command provides a snapshot of the IDS engine's current performance and configuration state on a specific host.

SSP monitors deployment status and generates platform-related alarms highlighting errors requiring attention. When allocated resources exceed predetermined threshold values, SSP generates an alarm to alert administrators. SSP instance health status appears as Yellow or Red when subcomponent issues require investigation. On the firewall side, an alarm is triggered when dropped threat events are detected and remains active until no dropped events are detected, keeping administrators informed of policy enforcement gaps. Malware Prevention service deployment issues surface as alarms in the NSX Manager UI. The License Hub dashboard highlights platform-related alarms displaying system alerts and critical errors requiring attention.

SSP monitoring tracks the status of the flow clustering job, which runs every hour, reporting status values including COMPLETED, FAILED, SUBMISSION_FAILED, FAILING, INVALIDATING, NOT_AVAILABLE, PENDING_RERUN, RUNNING,

and SUBMITTED. Security Intelligence suspicious traffic detectors report execution status after each run, with values of NOT_STARTED, SUCCESS, NOT_ENOUGH_BASELINE, or FAILURE.

The SSP Inventory monitoring capability displays the infrastructure hierarchy including regions, zones, environments, and virtual machines with operational status indicators that surface success, timeout, and out-of-sync conditions. Platform Inventory provides Power State information identifying whether a VM is running, stopped, suspended, or unknown. SSP displays health status for all onboarded Bare Metal servers, including BMS ID, BMS Name, Installer Version, Overall Status, and component-level details such as Control Connections and Distributed Firewall enablement status. The CLI provides a command to display the health of all services and agents running on a Bare Metal Server.

Network Detection and Response (NDR) can be integrated with SIEMs and generates event logs when detection events occur, campaigns are created, or campaigns are updated. SSP SIEM integration sends campaign and event logs to connected SIEM solutions including VCF Log Management and Splunk as events occur.

vDefend's monitoring scope is limited to its own security platform components. Centralized monitoring and alerting for the broader organizational Technology Assets, Applications, and Services inventory required by CAP-04 requires separate monitoring tooling outside of vDefend.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) provides automated monitoring and alerting capabilities scoped to disaster recovery and replication health, contributing to centralized visibility into the operating state of critical recovery infrastructure.

PNR monitors connections between protected and recovery sites by sending periodic ping requests to detect broken connections and logging connection check events. When connectivity issues are encountered, or when protection groups and recovery plans enter an error state, alarms are generated automatically. PNR also monitors resource consumption on the Protection and Recovery Server host and raises an alarm if a resource threshold is reached; both threshold values and alarm behavior are configurable. Alarms can be configured to send email notifications, SNMP traps, or run scripts on the vCenter host in response to events. For email notifications to function, Mail settings must be configured in the vCenter Settings menu and an administrator email address must be specified during appliance registration.

vSphere Replication generates informational, warning, and error events that enable operators to monitor replication system health and resolve potential problems. Administrators can define and edit alarms that trigger when specific replication events occur, such as when a virtual machine is configured for replication or enters an error state. PNR dashboards display a historical chart of bandwidth consumed by replication tasks relative to available capacity, as well as VM utilization trends over time to identify virtual machines at risk of performance degradation.

PNR provides default health alerts for protection groups, recovery plans, cloud file systems, recovery SDDCs, and protected sites through VCF Operations. The VCF Operations Management Pack for PNR tracks server pairing status, server connectivity, recovery plan execution states, recovery plan errors, and protection group status, integrating DR-specific health metrics into a broader centralized monitoring environment. In environments with more than one vCenter, PNR displays events from all Protection and Recovery servers registered as extensions. PNR also tracks recovery history metrics including error count, execution time, result state, warning count, and powered-on and powered-off VM counts, providing a structured view of recovery plan performance over time. The Clean Room Orchestrator includes a monitoring interface that displays the operational status of configurations related to recoverability and supportability.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) includes built-in performance monitoring and alerting capabilities for the database instances it manages and the infrastructure those instances depend on. Administrators and users can access monitoring data through the DSM Monitoring tab, accessible from both the vSphere Client plug-in and the standalone DSM UI. The Monitoring page displays real-time resource and performance metrics for each database VM, including Active Connections, Data Disk Usage, Write Throughput, Read Throughput, and CPU utilization. For MySQL instances, engine metrics are available including Slow Queries per Second, InnoDB Buffer Usage, InnoDB Reads and Writes, and Command Reads and Writes. A dedicated Monitoring tab at the individual database instance level provides per-database visibility into operating state and health status.

DSM generates alerts at three severity levels: OK, WARNING, and CRITICAL. Administrators and users can configure the threshold values that trigger each severity for the key health parameters: CPU Health, Data Disk Health, System Disk Health, and Max Connections. Separate warning and critical thresholds are configurable per database instance through the Manage Alerts for Databases interface in the DSM UI, which allows teams to tune alerting sensitivity to operational requirements. DSM generates named alert types such as CPU_HEALTH_ALERT when CPU utilization exceeds a configured threshold.

In addition to database-level monitoring, DSM surfaces infrastructure-level conditions through a Global Alerts page visible in both the plug-in and standalone interfaces. Infrastructure alerts cover conditions such as VCENTER_NOT_REACHABLE (when the Provider appliance cannot reach VMware vCenter via the configured FQDN), VCENTER_CONNECTIVITY (when vCenter connectivity fails), INFRASTRUCTURE_POLICY_VC_ALARMS (when active vCenter resource alarms are present), orphaned VM detection, vCenterBinding API status, vCenter service account creation failures, and license-related conditions. These infrastructure-level alerts are paired with continuous, granular performance monitoring so that administrators can correlate resource conditions with database service impact.

DSM supports configurable notification routing through Webhook settings, configured under Managing Settings in the DSM UI. Webhooks transmit alert notifications for data service health events and can be configured to include alert level information indicating whether an alert is CRITICAL or WARNING, enabling integration with existing operational tooling and notification systems.

DSM introduces monitoring integration with VCF Operations. VCF Operations automatically discovers and monitors PostgreSQL and MySQL clusters in the Ready state, displaying performance data across three levels: cluster-level for overall health, node-level for infrastructure resource use, and database-level for database-specific KPIs such as Active Connections, Commits, Rollbacks, and Deadlocks. PostgreSQL-specific engine metrics include Commits and Rollbacks, Deadlocks and Conflicts, and Tuples Updated. MySQL-specific engine metrics include Queries and Questions, Bytes Received and Sent, and Slow Queries. Active alerts specific to individual nodes or the overall cluster are visible within VCF Operations. PostgreSQL databases support the pg_stat_statements extension to track execution statistics across all SQL statements, enabling query-level performance analysis.

For SQL Server databases managed by DSM, the monitoring framework supports metrics-based and status-condition-based alerts to identify critical failure conditions including database engine degradation, non-operational status, and automated or transaction log backup failure.

DSM provides automated monitoring and alerting for the database platform tier it manages directly. It does not function as an enterprise-wide centralized monitoring platform aggregating data from all technology layers, so organizations typically integrate DSM alerting and VCF Operations visibility with an enterprise monitoring solution and define policies for alert triage, escalation, and response.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer (Avi) includes a built-in Analytics subsystem that collects and presents performance metrics for virtual services, Service Engines, and backend pool servers. The Avi Controller Metrics service gathers this performance data and visualizes it across the Platform dashboard. The Analytics tab for virtual services displays metrics including end-to-end timing, traffic patterns, logs, and events. Service Engine analytics provides historical charts of metric tiles for the current virtual service, pool, or Service Engine, with configurable time periods ranging from the past 30 minutes to the past year. Performance data is accessible through the Controller UI and via the REST API, which returns aggregated metrics at configurable intervals, enabling external monitoring systems to ingest Avi operational data programmatically. The Avi for VCF guide confirms that Virtual Service Analytics captures and reports logs and events for application performance monitoring in VCF deployments. Within API gateway use cases, the monitoring capability tracks complete API transaction logs including response time and end-user experience data.

Avi supports configurable metric-based and event-based alerting. The Alert feature processes system events and metrics against user-defined rules and generates alerts that initiate configured actions. Alert actions deliver notifications through five channels: the local Controller UI, email, syslog, SNMP traps, and ControlScript for custom automated responses. SNMP notifications support both SNMPv2c and SNMPv3 protocols, and can be configured with TLS encryption and strict certificate

verification for secure transmission to external trap receivers. Avi supports multiple SNMP servers for alert notifications. These notification destinations are configured in the Operations > Notifications page of the Controller. Custom metric-based alerts can be defined against specific thresholds. Status-change events, such as Virtual Service Down events, generate SNMP trap notifications automatically. Autoscale alert actions can also generate informational notifications via email, syslog, or SNMP traps when Service Engine scaling occurs in response to load changes. The Avi Load Balancer Metrics Scripted Input pulls performance metrics using the REST API, supporting integration with Splunk for centralized collection of performance data.

Avi Service Engine monitors and alerts on resource utilization thresholds including CPU usage, memory usage, disk usage, and packet buffer usage. Detailed per-process CPU metrics are available, covering process name, CPU usage percentage, and memory usage percentage for Service Engine processes. Authorized users can also connect directly to a Service Engine and run standard CPU utilization monitoring tools to observe performance metrics in real time.

Avi health monitors continuously check the operating state of backend pool servers by issuing health-check requests to each pool member. Health monitors can detect when a server has entered maintenance mode by evaluating specific response codes or response data returned by the server, allowing Avi to route traffic away from pool members in maintenance. HTTP, TCP, and UDP health monitors all support maintenance mode detection via configurable response strings or response codes using the `maintenance_response` and `maintenance_code` parameters.

The Avi Controller exposes its own operational metrics through a REST API endpoint, including CPU usage statistics and process-level performance data. This enables an external monitoring platform to collect and centralize health and performance data from the Avi management plane alongside data from other infrastructure components.

